Supplementary Material

EgoCAMRA: Egocentric Confidence-Aware Residual Random Walks for Multi-Action Recognition and Anticipation

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In this supplementary material, we provide detailed description of the datasets and additional qualitative results omitted in the main paper.

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A. Detailed Information on the Experimented Datasets

1) **EPIC-Kitchens [1]:** The EPIC-Kitchens Dataset consists of egocentric videos of day-to-day activities carried out by unique subjects inside their kitchens. This data set includes 432 videos shot by 32 unique individuals carrying out non-scripted kitchen activities. These videos have been shot with a GoPro camera strapped onto the subject's head in such a way that the view can be altered based on the kitchen environment. The dataset is annotated for 352 object categories and 125 action classes, providing a rich foundation for studying complex hand–object interactions and multimodal learning in egocentric scenarios. In total, it includes more than 28K annotated video segments covering a wide variety of cooking and interaction sequences. All videos are provided in MPEG-1 format with a spatial resolution of 1920×1080 at 59.94 fps. Owing to its scale, diversity, and naturalistic recordings, EPIC-Kitchens serves as one of the largest and most comprehensive benchmarks for egocentric action recognition and temporal activity understanding.

(a) **(b)** **(c)** **(d)**

Fig. 1: Sample frames from the EPIC-Kitchens dataset [1]

2) **ADL Dataset [2]:** The ADL (Activities of Daily Living) dataset is a collection of 20 first-person video streams, where each video lasts about an hour, summing up to a total of 10 hours. The videos are shot at a frame rate of 30 fps and have a resolution of 1280×720. They consist of 32 activity types and more than 40 different object categories. The activities are related to daily routines such as eating, reading, and object manipulation. The dataset offers dense annotations that include starting and ending frames of activities as well as object-level information. Due to continuous recording, ego-motion, and occlusions from hand movements, the dataset is challenging and well-suited for evaluating egocentric action recognition and activity understanding models.

(a) **(b)** **(c)** **(d)**

Fig. 2: Sample frames from the ADL dataset [2]

3) **Charades Dataset [3]:** Charades is an example of a large-scale dataset used to evaluate activity recognition tasks in realistic indoor scenes. This dataset contains 9,848 videos that were captured by crowd-sourcing workers in various home settings. In total, this dataset contains about 82 hours of videos with a frame rate of 30 fps and resolution of 1920×1080. The Charades video dataset contains a variety of scene changes, illumination conditions, and

human actions. In addition, there are 157 action labels related to the interactions between 46 objects, with multiple actions taking place simultaneously or consecutively. Moreover, the dataset provides dense annotations that allow determining the beginning and ending frames for each individual action. The dataset is considered complex because of the presence of overlapping actions, as well as realistic conditions, similar to Hollywood.

(a) **(b)** **(c)** **(d)**

Fig. 3: Sample frames from the Charades dataset [3]

B. Additional Ablation Studies on Charades Dataset

We report the ablation study results on the Charades dataset [3] to complement the results presented on EPIC-Kitchens and ADL in the main manuscript. All settings are identical to those described in the main paper.

Ablation Study 1: Effectiveness of Proposed Model Variants on Charades Dataset [3]

In the first ablation study, we show improvements when successively adding elements to our proposed EgoCAMRA framework on the Charades benchmark. The $\mathbf{B}+\mathbf{D}$ configuration, using RGB-Optical flow fusion along with the classifier, represents the lower bound since only these two elements are used in it. The addition of the EOAFE model element ($\mathbf{A}+\mathbf{B}+\mathbf{D}$) achieves an improvement in performance through the use of hand-object-centric features through ROI Align. It is proven that taking into account object-aware egocentric cues is important. The addition of the graph construction component ($\mathbf{A}+\mathbf{B}+\mathbf{C}+\mathbf{D}$) leads to steady performance gains due to the consideration of temporal relations between frames. The mean accuracy values in Table I clearly show that the full model $\mathbf{A}+\mathbf{B}+\mathbf{C}+\mathbf{D}+\mathbf{E}$ achieves the best mAP of 0.6294, consistent with the trends observed on EPIC-Kitchens and ADL, confirming that each module plays a complementary and necessary role across all three datasets.

TABLE I: Effectiveness of Proposed Model Variants on Charades Dataset [3].

Variant	mAP
$\mathbf{B}+\mathbf{D}$ (Baseline)	0.4618
$\mathbf{A}+\mathbf{B}+\mathbf{D}$	0.5218
$\mathbf{A}+\mathbf{B}+\mathbf{C}+\mathbf{D}$	0.5580
$\mathbf{B}+\mathbf{C}+\mathbf{D}+\mathbf{E}$	0.5898
$\mathbf{A}+\mathbf{B}+\mathbf{C}+\mathbf{D}+\mathbf{E}$	**0.6294**

*($\mathbf{A}$ = EOAFE, $\mathbf{B}$ = RGB + Optical Flow, $\mathbf{C}$ = Graph, $\mathbf{D}$ = Classifier, $\mathbf{E}$ = Residual Random Walk.)

Ablation Study 2: Effectiveness of Different Feature Extraction Backbones on Charades Dataset [3].

In the second ablation study, we investigate the effect of different feature extraction backbones and input modality combinations on recognition performance on the Charades dataset, as reported in Table II. We compare EfficientNet-B0 ($\mathcal{E}_{Net}$), VGG16 ($\mathcal{V}_{Net}$), and ResNet-50 ($\mathcal{R}_{Net}$) across RGB ($\mathbf{F}_{\{rgb\}}$) and Optical Flow ($\mathbf{F}_{\{flow\}}$) modalities. Using only $\mathbf{F}_{\{flow\}}$ results in the lowest mAP of 0.5200, suggesting that optical flow alone lacks sufficient discriminative power for multi-label action recognition on Charades. The mean accuracy values in Table II clearly show that $\mathcal{R}_{Net}$ with full modality fusion achieves the best mAP of 0.6294, consistent with the findings on EPIC-Kitchens and ADL, making it the backbone of choice for the proposed framework.

TABLE II: Effectiveness of Different Feature Extraction Backbones on Charades Dataset [3].

Modality	Backbone	mAP
$\mathbf{F}_{\{rgb,flow\}}$	$\mathcal{E}_{Net}$	0.6107
$\mathbf{F}_{\{rgb,flow\}}$	$\mathcal{V}_{Net}$	0.6119
$\mathbf{F}_{\{rgb\}}$	$\mathcal{R}_{Net}$	0.6081
$\mathbf{F}_{\{flow\}}$	$\mathcal{R}_{Net}$	0.5200
$\mathbf{F}_{\{rgb,flow\}}$	$\mathcal{R}_{Net}$	**0.6294**

Ablation Study 3: Effectiveness of Loss Function on Charades Dataset [3].

In the third ablation study, we examine the contribution of each loss function component on the Charades dataset, as reported in Table III. Using only the classification loss $\zeta_1 = \mathcal{L}_{cls}^{weighted}$ serves as the baseline. The inclusion of graph smoothness loss ($\zeta_1 + \zeta_2$) always leads to better results, which proves that graph smoothness contributes to learning more informative features. Likewise, $\zeta_1+\zeta_3$ also leads to considerable improvement, which indicates that temporal coherence is beneficial as supervisory information. The average accuracies obtained for each case shown in Table III demonstrate that the combination of $\zeta_1 + \zeta_2 + \zeta_3$ produces the best mAP result of 0.6294.

TABLE III: Loss Function Comparison on Charades [3].

Loss Function	mAP
ζ_1	0.4500
$\zeta_1 + \zeta_2$	0.5900
$\zeta_1 + \zeta_3$	0.5877
$\zeta_1 + \zeta_2 + \zeta_3$	**0.6294**

Ablation Study 4: Effectiveness of RW vs. CARRW on Charades Dataset [3].

In the fourth ablation study, we compare the classic random walk ($\mathcal{RW}$) against the suggested confidence-aware residual random walk ($\mathcal{CARRW}$) on the Charades dataset, as illustrated in Table IV. Unlike the classic approach, $\mathcal{CARRW}$ has a residual component that preserves the first prediction probability distribution in every step of the process. The confidence-aware weighing mechanism guarantees that only those frames predicted with high confidence influence the propagating process, preventing over-smoothing. As can be seen from Table IV, $\mathcal{CARRW}$ performs better than $\mathcal{RW}$, with an increase in the average precision metric by $+0.0394$.

TABLE IV: RW vs. CARRW on Charades Dataset [3].

Method	mAP
$\mathcal{RW}$	0.5900
$\mathcal{CARRW}$	**0.6294**

$\mathcal{RW}$ = Random Walk, $\mathcal{CARRW}$ = Confidence-Aware Residual Random Walk.

C. Additional Qualitative Results for Multi-action Recognition and Anticipation

This section presents additional qualitative results complementing those reported in the main manuscript, including co-occurrence timeline analysis on EPIC-Kitchens, feature space visualization, and transition matrix visualization at different anticipation horizons.

Co-occurrence Timeline — EPIC-Kitchens

Figure 4 presents the multi-action co-occurrence timeline for EPIC-Kitchens (P01_04), complementing the ADL results shown in the main manuscript. Each horizontal bar represents a single action segment, and the red-shaded regions highlight temporal intervals where multiple actions overlap simultaneously. From the figure, it can be clearly seen that the proposed model correctly captures simultaneous action overlaps, and the lower panel confirms that CARRW successfully assigns high confidence scores to co-occurring actions at the correct temporal locations, demonstrating the model's ability to handle complex egocentric action scenarios involving overlapping activities.

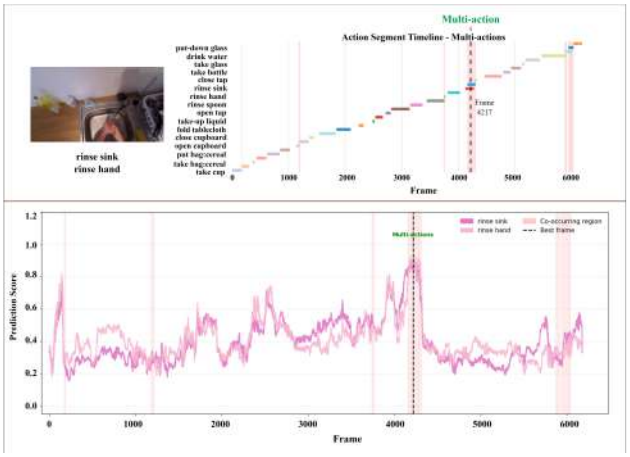

Fig. 4: Multi-action co-occurrence timeline for EPIC-Kitchens (P01_04). Pink-shaded regions indicate temporal intervals of simultaneous action overlap, and the lower panel confirms that CARRW assigns high confidence scores to co-occurring actions at the correct temporal locations.

Feature Space Visualization

Figure 5 presents the feature space evolution from the input representation to the learned feature space, along with the corresponding class-wise confidence scores on EPIC-Kitchens (P01_02). The top row shows the t-SNE visualization of input features, while the bottom row

shows the learned feature space after CARRW propagation. As can be seen from the illustration, target action clustering becomes increasingly tighter in the learned feature space relative to the input space, whereas CARRW boosts the confidence for multiple target actions while suppressing the non-target ones, illustrating its ability for learning multiple action relationships effectively.

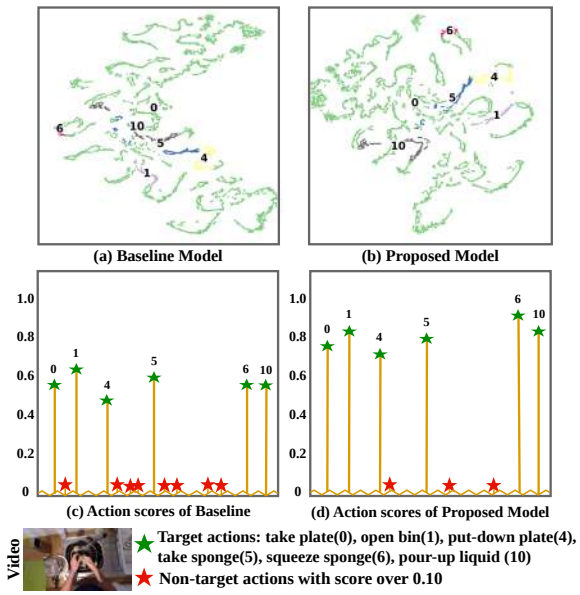

Fig. 5: Feature evolution from input to learned feature space and corresponding action scores for multi-action relationships on EPIC-Kitchens (P01_02).

Transition Matrix Visualization

Figure 6 visualizes the learned transition matrix $\mathbf{P}^{\nabla}$ at horizons $\nabla = 15, 30, 45, 60$ frames, corresponding to 0.25s, 0.50s, 0.75s, and 1.00s respectively. From the figure, it can be clearly seen that at $\nabla = 15$ (0.25s), a strong diagonal structure confirms that temporally adjacent frames carry the highest mutual influence. As ∇ increases, transition weights gradually diffuse to a wider temporal neighborhood, reflecting the natural uncertainty that accumulates with increasing anticipation horizon. Importantly, the weights remain structured and non-uniform even at $\nabla = 60$ (1.00s), confirming that $\mathbf{P}^{\nabla}$ retains reliable temporal relationships across all horizons without requiring any additional learned parameters.

D. Failure Case Analysis

To provide a complete evaluation of the proposed framework, we present failure cases observed on the EPIC-Kitchens dataset. These cases highlight scenarios where the proposed model produces incorrect predictions, providing insights into the current limitations of the framework.

Failure Cases on EPIC-Kitchens

The proposed framework encounters challenges in two main scenarios as shown in Figure 7. The first limitation is that, with rapidly repeated actions, the STSG may have a tendency to provide a large similarity value between the irrelevant frames, resulting in a failure to predict

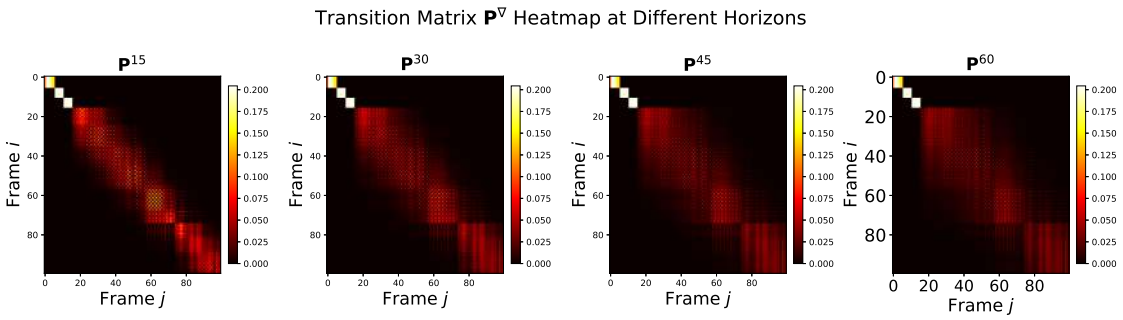

Fig. 6: Visualization of $\mathbf{P}^{\nabla}$ at horizons $\nabla = 15, 30, 45, 60$ (0.25s, 0.50s, 0.75s, 1.00s). The diagonal structure confirms temporal proximity modeling, while gradual weight diffusion reflects natural temporal uncertainty accumulation.

the labels during CARRW. The second limitation is related to the low number of training samples for the rare action classes. In this case, the weighting scheme based on the confidence measure may excessively suppress the predictions, causing missing detections.

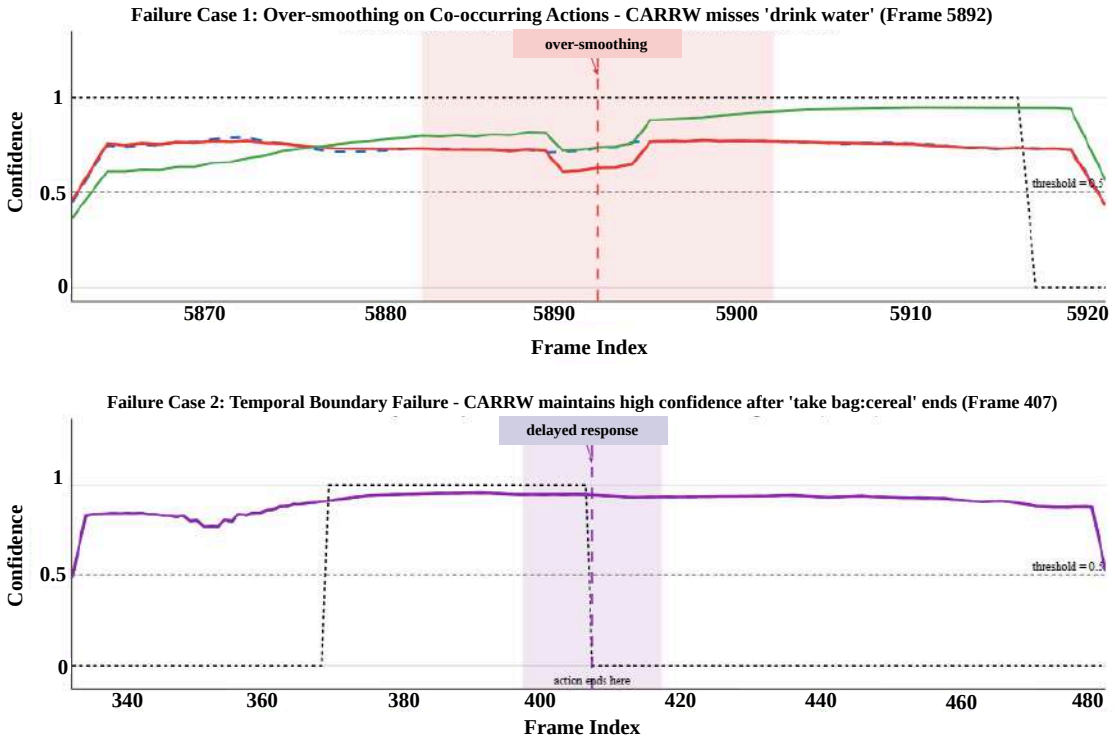

Fig. 7: Failure cases of CARRW on EPIC-Kitchens (P01_04). (Top) Over-smoothing on co-occurring actions: at frame 5892, CARRW misses *drink water* while correctly predicting the concurrent *put-down bottle*. (Bottom) Temporal boundary failure: CARRW holds a high confidence for *take bag:cereal* past its true end at frame 407, yielding a delayed response. Dashed black lines denote ground truth; the horizontal dotted line marks the 0.5 detection threshold.

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